

Nonnegative and positive factorizations

Complexity of globally optimal nonnegative rank-two approximation

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Rating rationale: Challenging because rank two still requires a global complexity classification despite tractable special inputs; community importance comes from the basic gap between SVD approximation and constrained NMF.

Topic: low-rank approximation; computational complexity

Resolution — 2026-09-11

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review:** PASS for the exact target.

The exact rational-input decision problem is NP-hard under polynomial-time many-one reductions, even for strictly positive symmetric positive-definite inputs. The proof supplies an inverse-polynomial additive squared-error gap and a polynomial-time rational perturbation to simple spectrum. Factors may be real, exactly as in the canonical question. NP membership and constant-relative-error hardness are not asserted.

The complete target is resolved. Its former difficulty rating is historical; the original statement, references and dated audits remain below.

Primary reference: [complete authored PDF](#), [standalone TeX](#), [Theorem 1](#); [Theorem 5 and Corollary 6 strengthen the construction](#). [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Problem statement

Given an arbitrary $X \in \mathbb{Q}_+^{m \times n}$ and $\tau \in \mathbb{Q}_{\geq 0}$, determine the complexity of deciding whether

$$\exists W \in \mathbb{R}_+^{m \times 2}, H \in \mathbb{R}_+^{2 \times n} : \sum_{i=1}^m \sum_{j=1}^n \left(X_{ij} - \sum_{k=1}^2 W_{ik} H_{kj} \right)^2 \leq \tau.$$

In particular, is there a deterministic algorithm polynomial in the total binary input length, or is this decision problem NP-hard under polynomial-time many-one reductions? This is a complexity-classification question, without an assumption that these two outcomes exhaust the possibilities. The factors may have real entries; only the data and threshold must be rational. Their inner dimension is at most two, with a zero factor column permitted.

This fixes an exact decision interpretation of the literature's global rank-two NMF optimization question. There is no promise that X itself has rank two. When a rank-two truncated SVD of X is nonnegative, a best nonnegative rank-two approximation is obtainable from it. Arbitrary input can fall outside this tractable special case, and alternating nonnegative least squares need not find a global optimum.

References

Gillis, *Nonnegative Matrix Factorization*, §6.1.3, p. 199, Theorem 6.6 and final paragraph. Lindy, Noferini, and Van Dooren, *On rank-2 Nonnegative Matrix Factorizations and their variants*, §1, with §3's suboptimal approximation and §4's ANLS initialization.

Status check — 2026-09-10

Rechecked the [Lindy–Noferini–Van Dooren preprint](#) and [Noferini's December 2025 seminar](#), which explicitly retains the rank-two complexity question. Inputs having a nonnegative best rank-two SVD approximation form a proved tractable subclass. Searches also found [Gouveia–Wiebe's 2026 integer-factor problem](#); its integer factors and exact-rank input are different. The [2026 constrained Gram-feasibility hardness result](#) also has additional affine entry constraints and is not this unconstrained approximation problem. No unrestricted rational-input decision classification was located.